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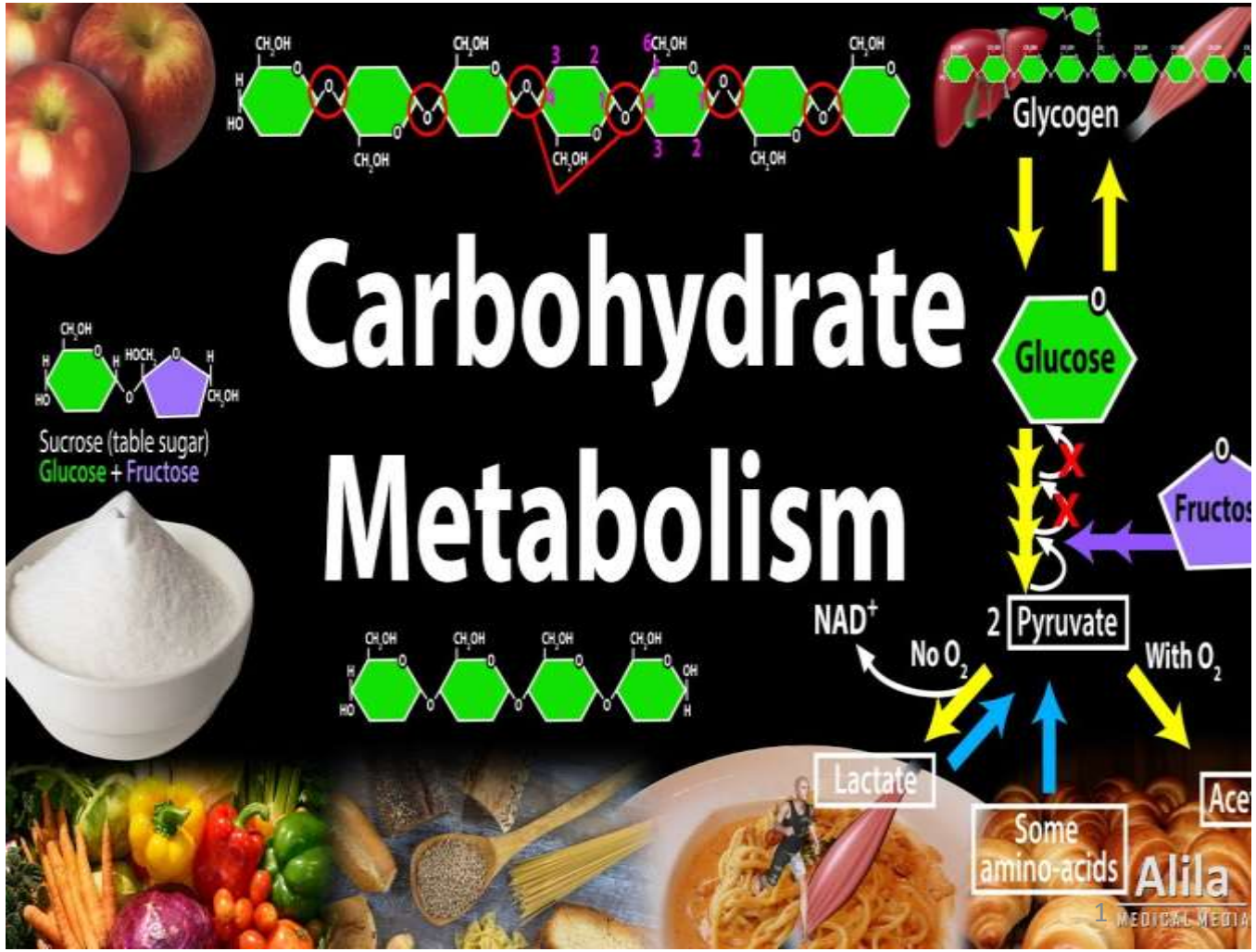
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Medicine College/HMU

2nd Stage

2026-2027



Metabolism

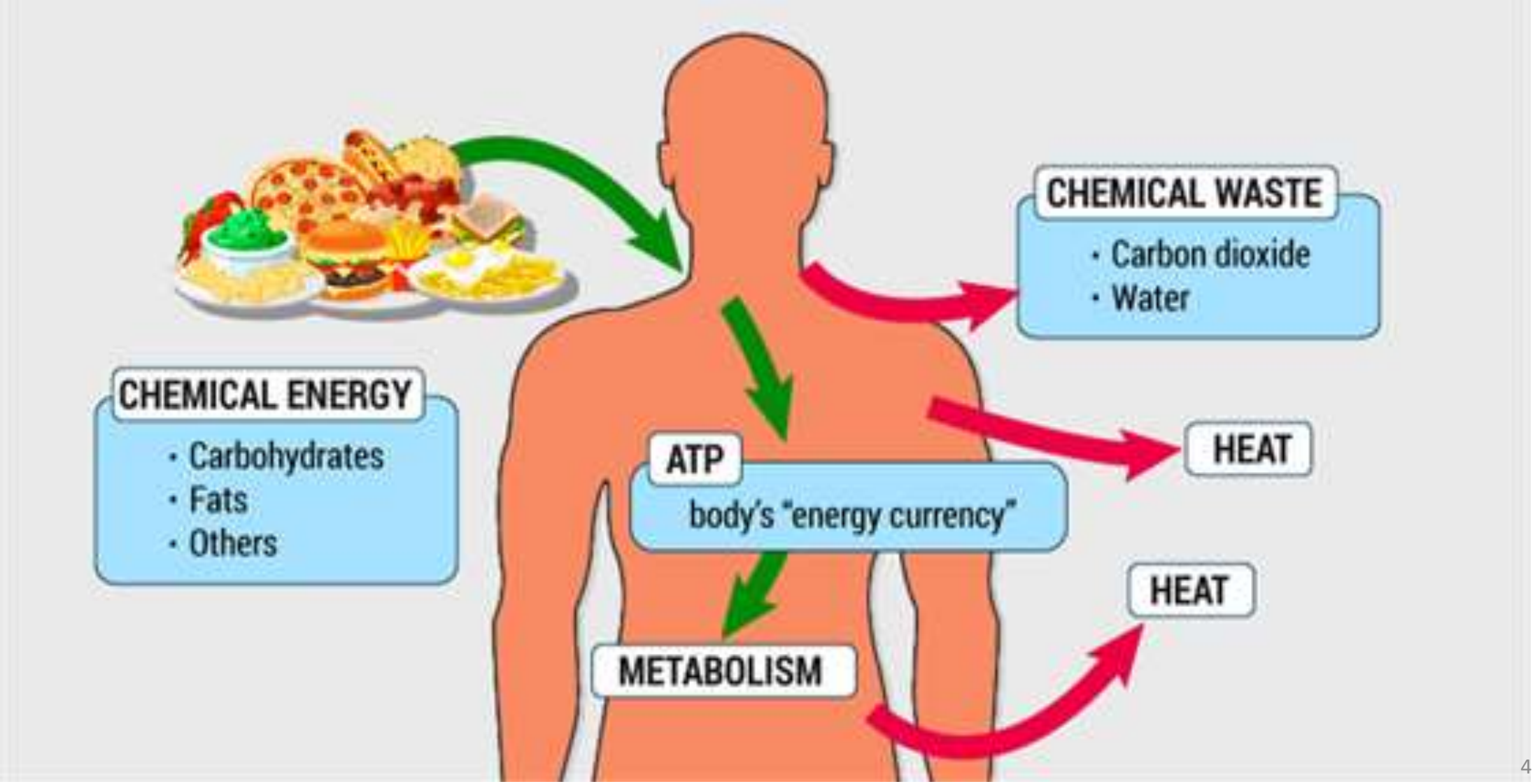
- Metabolism refers to the whole sum of reactions that occur throughout the body within each cell and that provide the body with energy.

- All living organisms require energy for different essential processes and for producing new organic substances.

- The chemical reactions by which metabolism occurs are almost the same in all living organisms, including animals, plants, bacteria, and fungi.

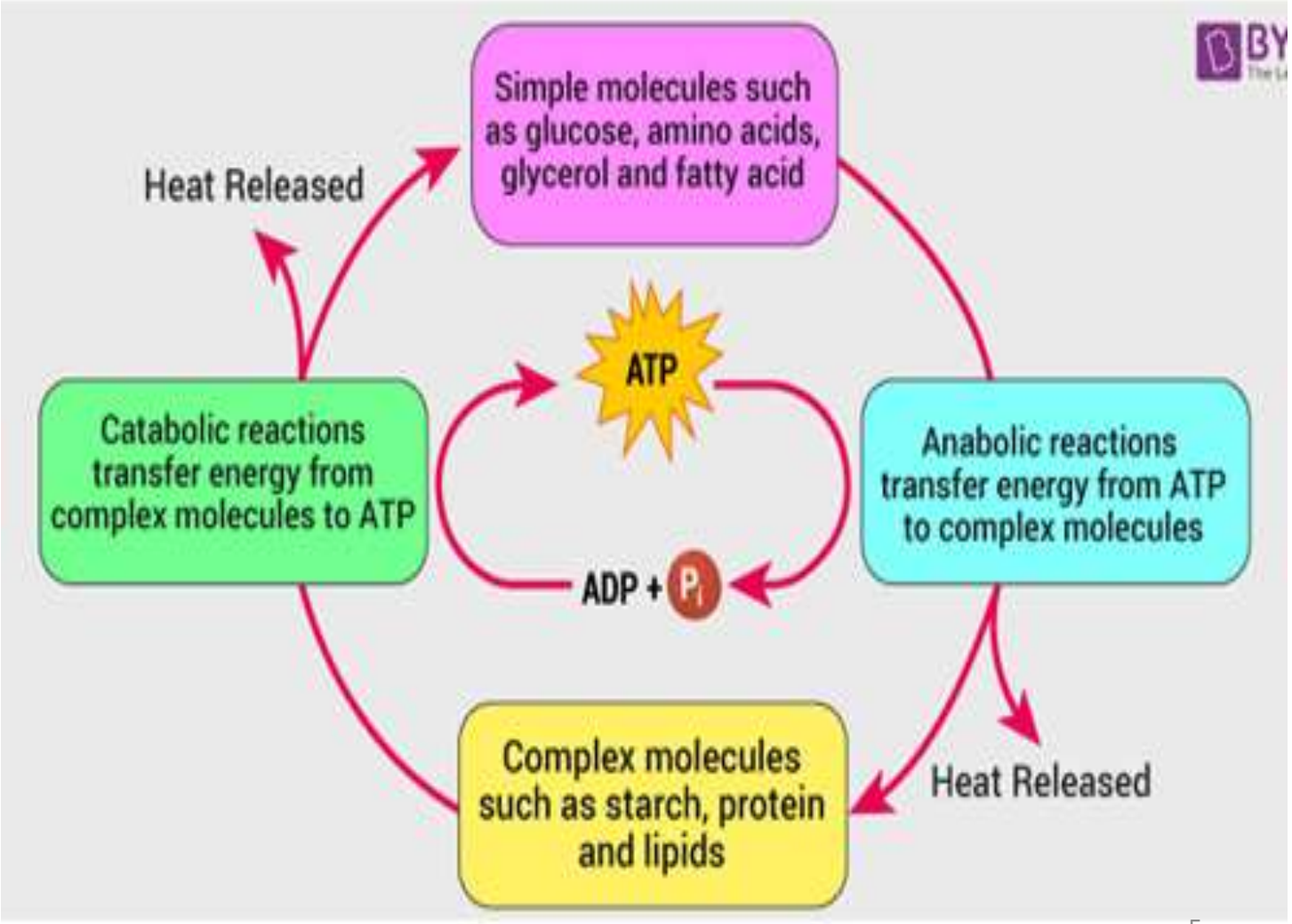
- All these chemical reactions are mediated by enzymes under specific environmental conditions such as pH and temperature.

Energy and Human Life



There are two types of metabolic process:

- Catabolism
- Anabolism

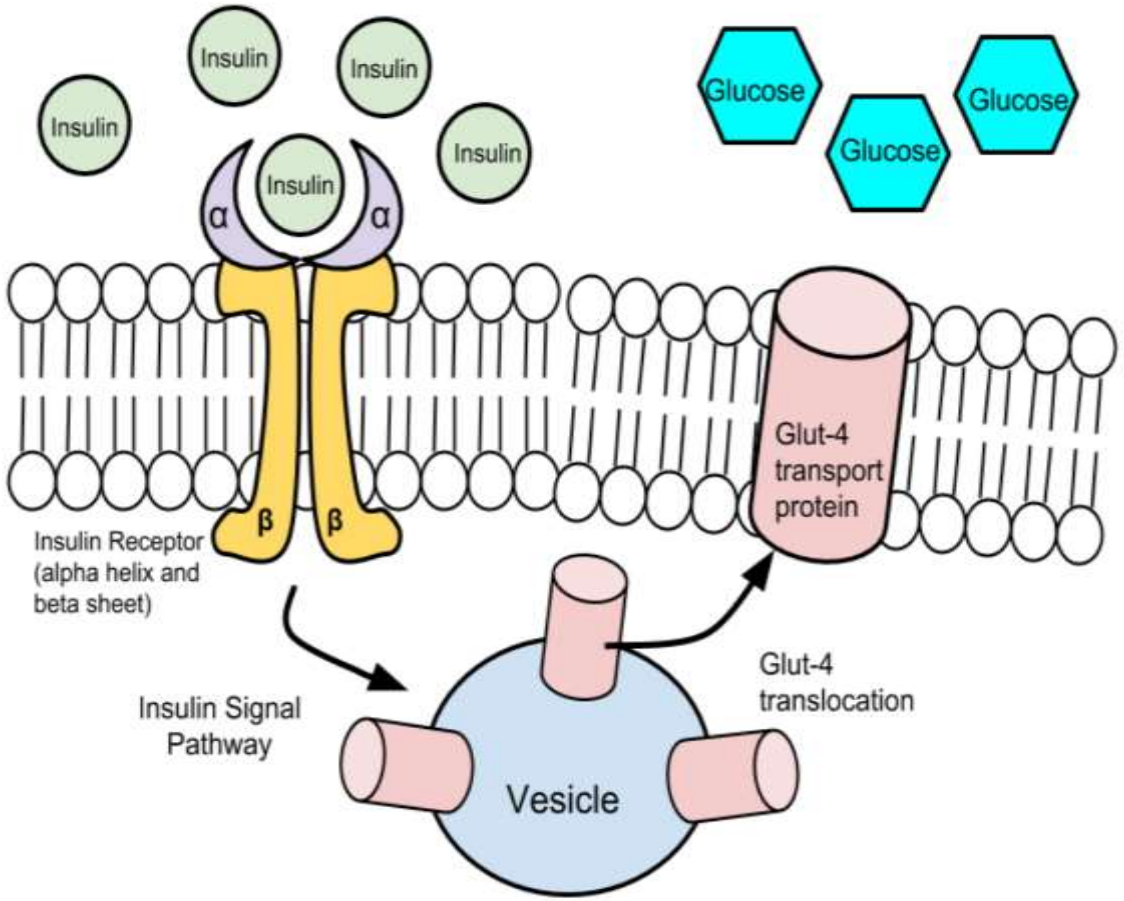


Carbohydrates in Metabolism

- Carbohydrates are supplied in three forms:
  - Starch
  - Sugar
  - Cellulose
- Starch and sugar are the major forms of energy for human.

How glucose enter the cell?

- Insulin binds to its receptor on the cell membrane.
- This activates an intracellular signaling pathway.
- The signal causes GLUT-4- containing vesicles to move toward the cell membrane.
- The vesicles fuse with the membrane, placing GLUT-4 on the cell surface.
- GLUT-4 allows glucose to enter the cell.



When glycolysis activated or inhibited?

Activated, when:

- Increase ATP demand
- Decrease ATP level
- Increase AMP/ADP
- Increase glucose availability
- Insulin is high (particularly in the liver & adipose tissue).
- Exercise (especially in skeletal muscle).

Inhibited, when:

- Increase ATP
- Increase Citrate (indicates sufficient energy).
- In the liver, fasting/low insulin and glucagon suppress glycolysis.

The metabolism of CHO subdivided into the following categories:

1- Glycolysis: (glycose: a term for glucose + -lysis degradation):

- It is the metabolic pathway that converts glucose $C_6H_{12}O_6$, into pyruvate.
- The free energy released in this process is used to form the ATP and NADH.

- Glycolysis is an oxygen independent metabolic pathway (anaerobic).

  - It can occur with or without oxygen.

- It occurs in the cytosol (cytoplasm) of almost all body cells.

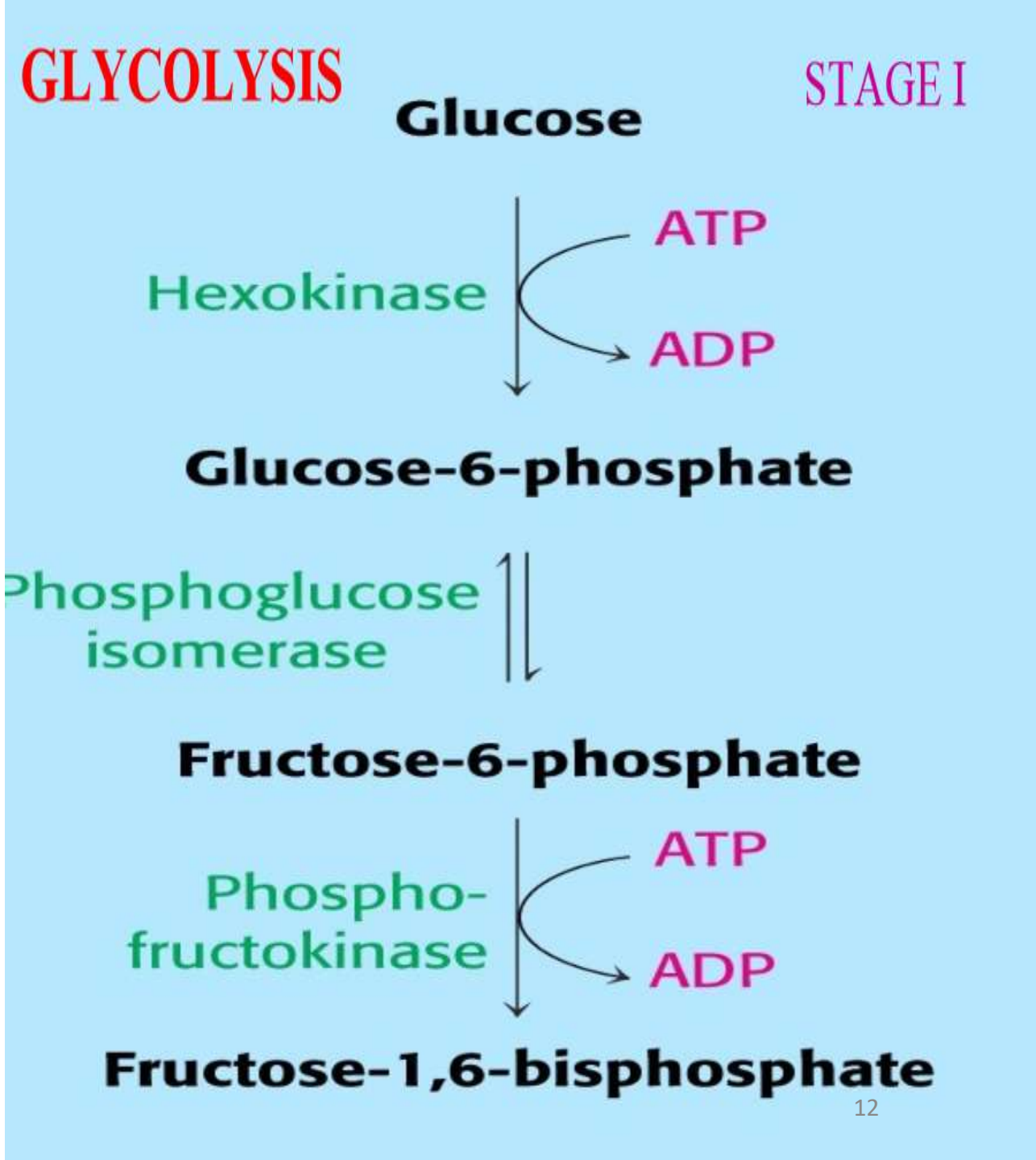
The overall process of glycolysis is:

$$Glucose + 2NAD^{+} + 2ADP + 2Pi \rightarrow 2Pyruvate + 2NADH + 2H^{+} + 2ATP + 2H2O$$

This process occur in the cytoplasm by the action of two enzymes.

- specific enzyme (glucokinase) in the liver.

- nonspecific enzyme (hexokinase), present in liver and other extra hepatic cell.



- The glucose in the blood circulation, when enter the cell become phosphorylated into (G-6-p).

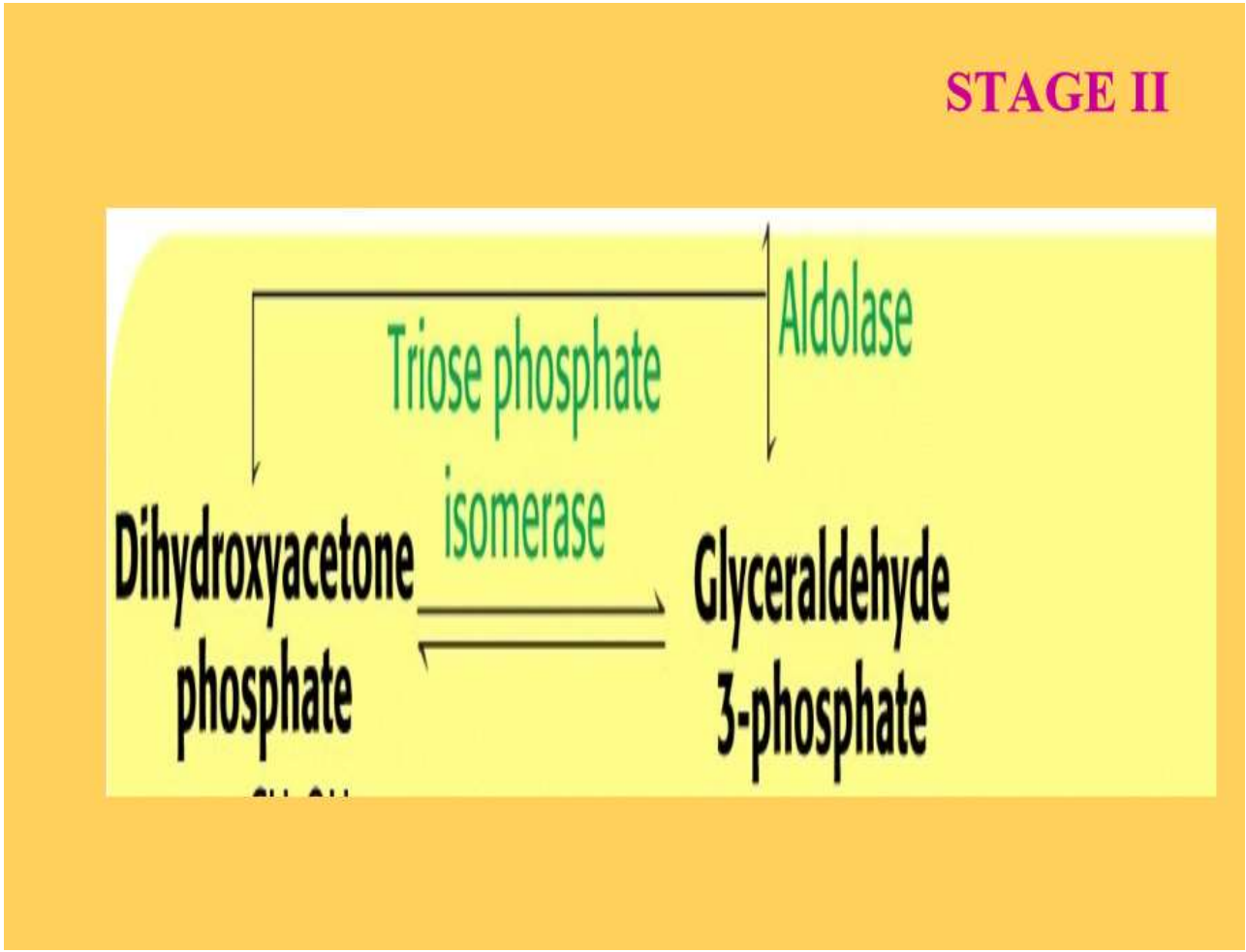
- Glu-6- p is an important compound for several metabolic pathways, and the reaction is irreversible.

- This stage consumes two ATP.

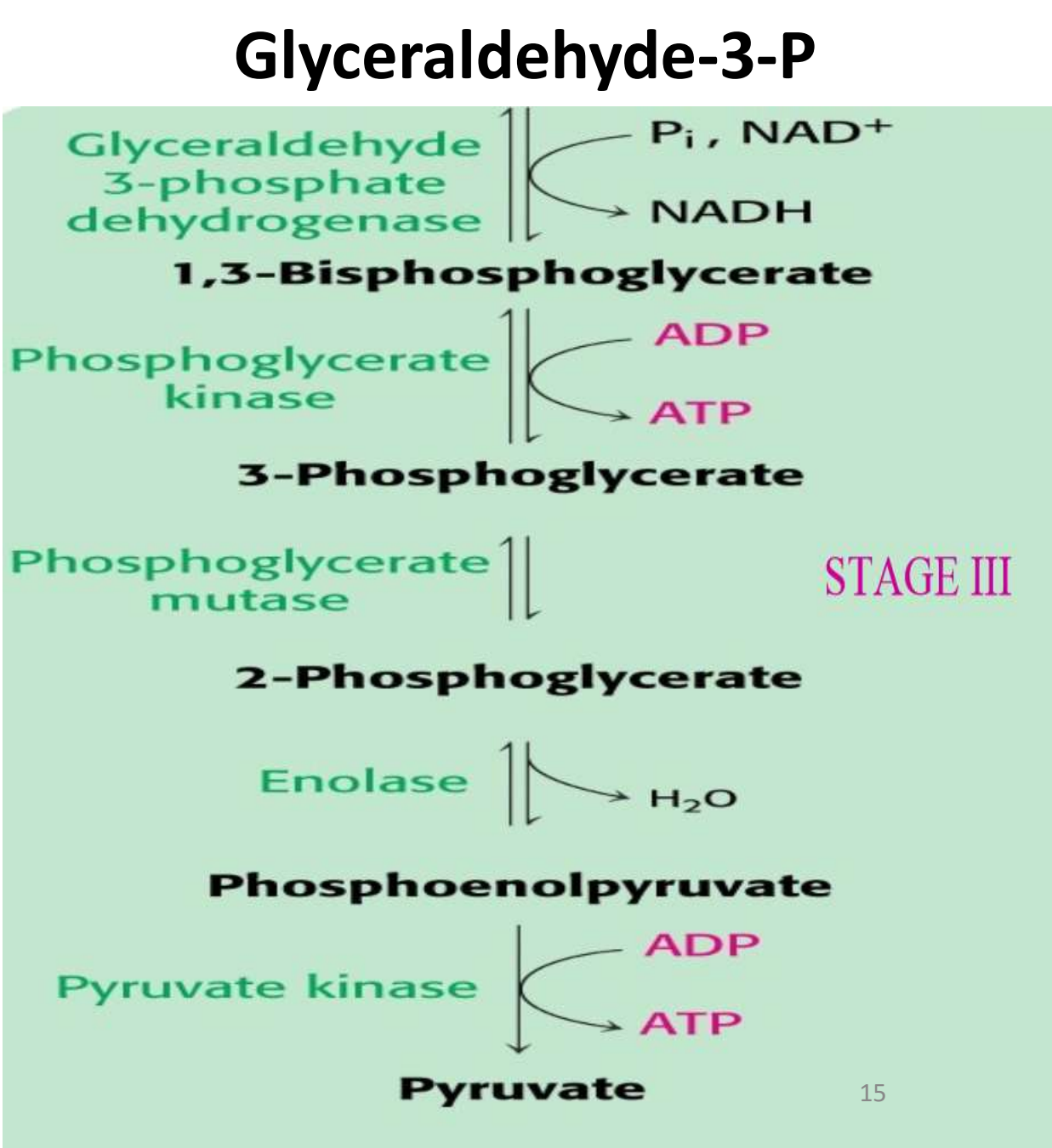
- By aldolase enzyme Fructose-1,6-diphosphate convert into two trioses (isomers).

- The two trioses interchangeable by a specific isomerase enzyme.

Fructose-1,6-bisphosphate



- Glyceraldehyde-3-P, through several steps, is converted into Pyruvate.
- 2ATP with 1NADH is produce.
- For 2Glyceraldehyde-3-p, the net energy become 4ATP+ 2NADH.



Energy production:

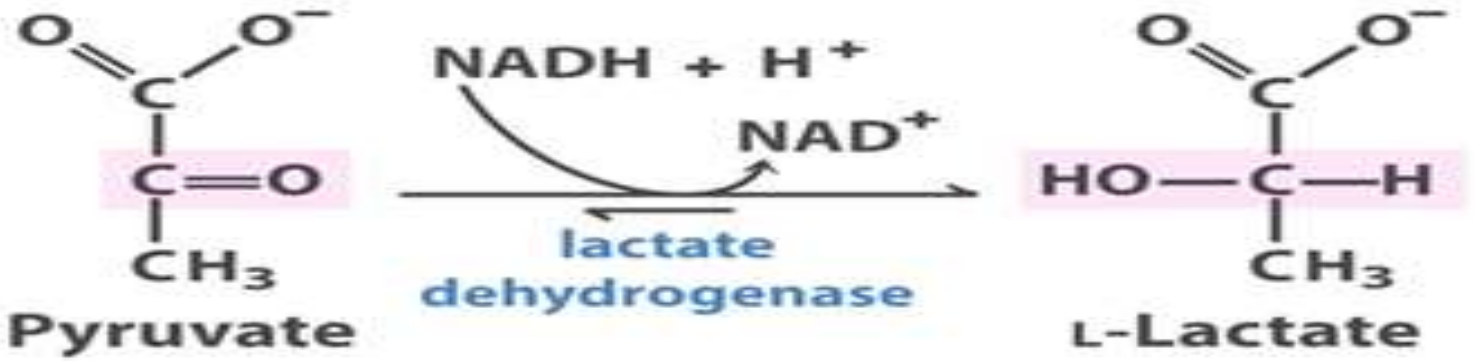
- The net of ATP molecules during glycolysis is equal to (8).
- ATP used
  * (glu → glu-6-pho) -1 ATP
  * (F-6-pho → F-1,6-dipho) -1 ATP

- ATP gain / Glucose molecule
  * Gly-3-pho →1,3-diphoGly +1NADH × 2
  * 1,3-diphoGly →3phoGly +1ATP × 2
  * PEP→ pyruvate +1ATP × 2

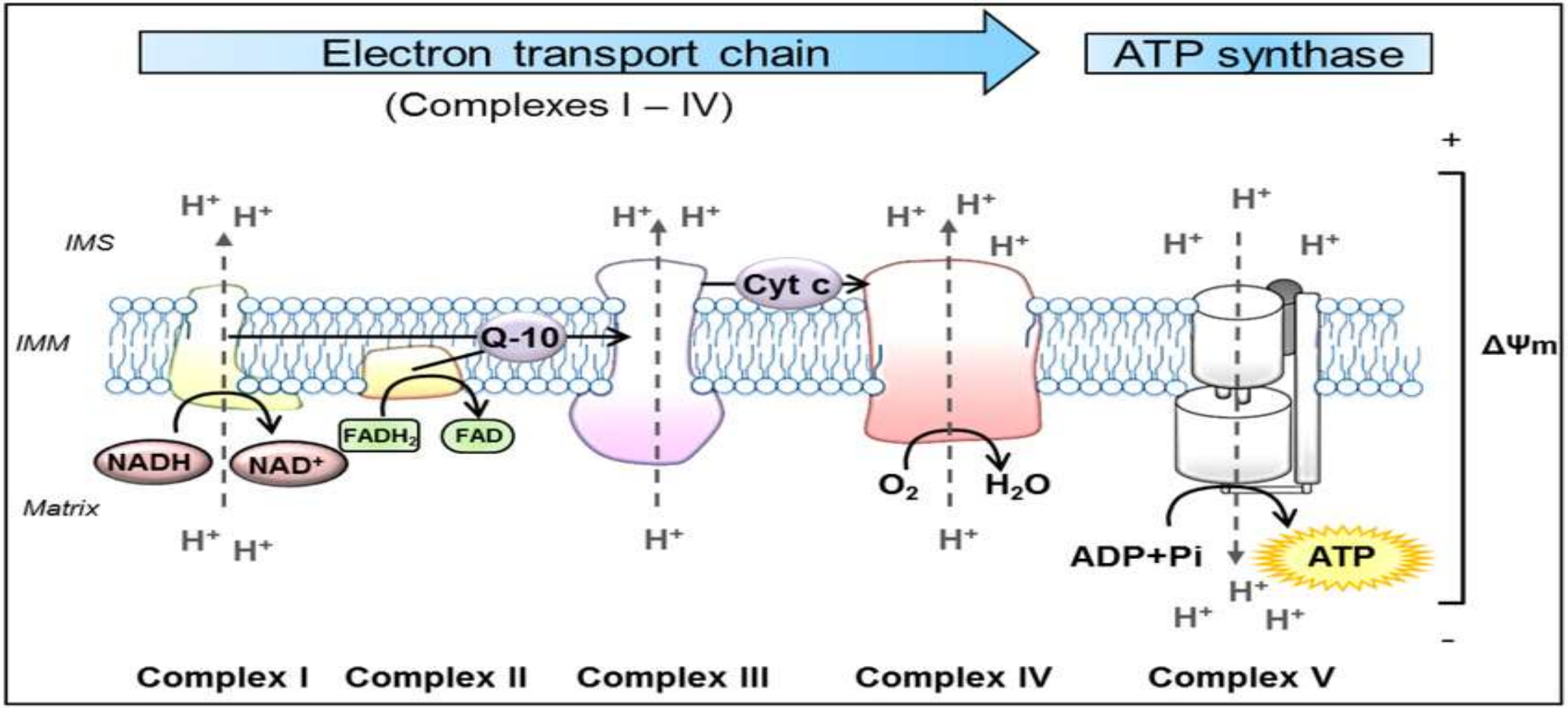
[(1NADH=3ATP) × 2 + 2ATP × 2] =

(6+4)ATP – 2ATP= 8 ATP

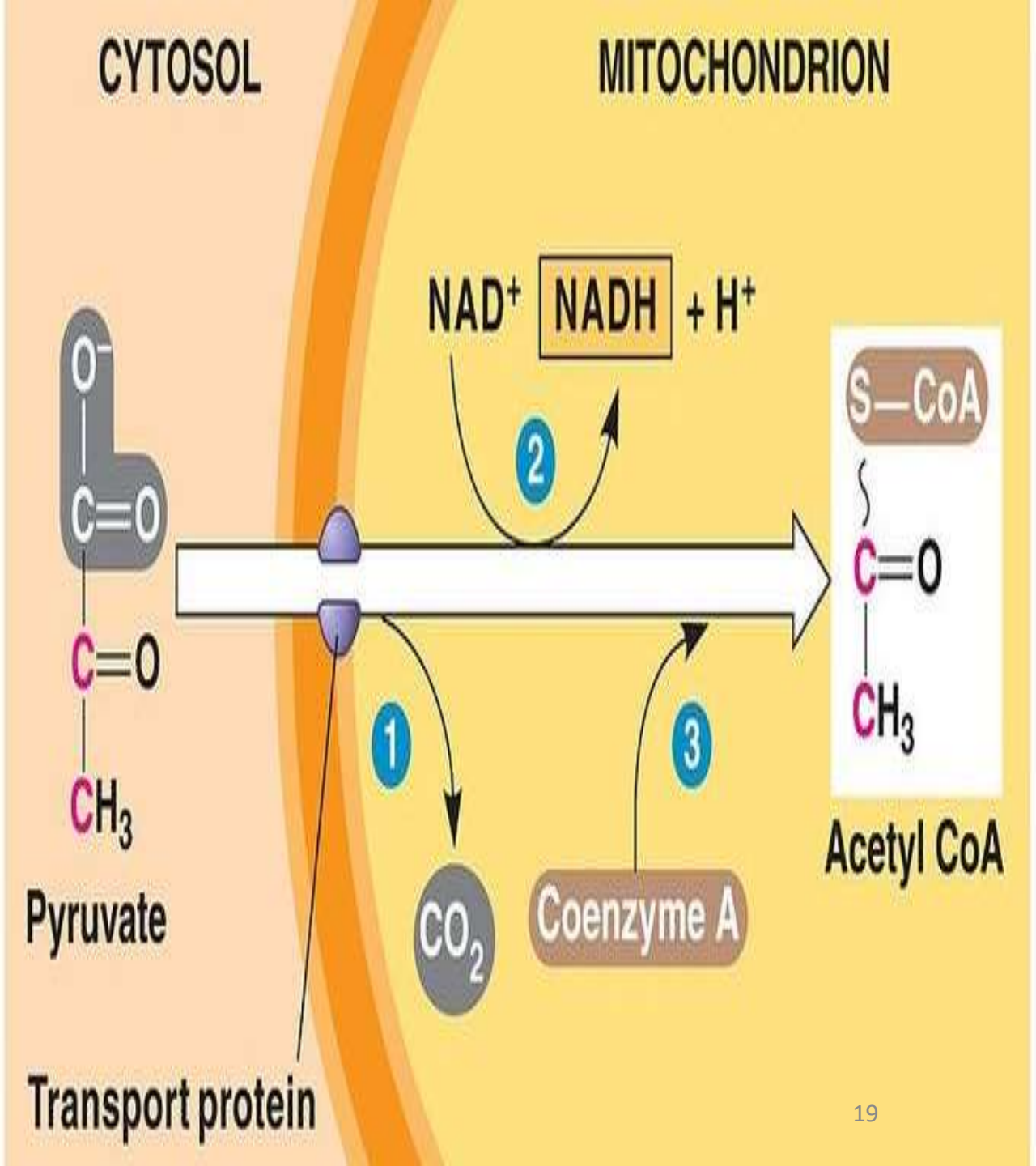
- Net of ATP from aerobic glycolysis (or from glucose to pyruvate) is 8 ATP.
- Formation of lactate from pyruvate is the major step in RBCs, lens and cornea, kidney, medulla, and leukocytes, resulting 2ATP (anaerobic glycolysis)! How?



Oxidative Phosphorylation



- Pyruvate is the bridge between glycolysis and the Krebs cycle,
- It is produced in glycolysis, and then is brought into the mitochondria where it forms acetyl Coenzyme A.
- Pyruvate enters the mitochondria to begin aerobic respiration.



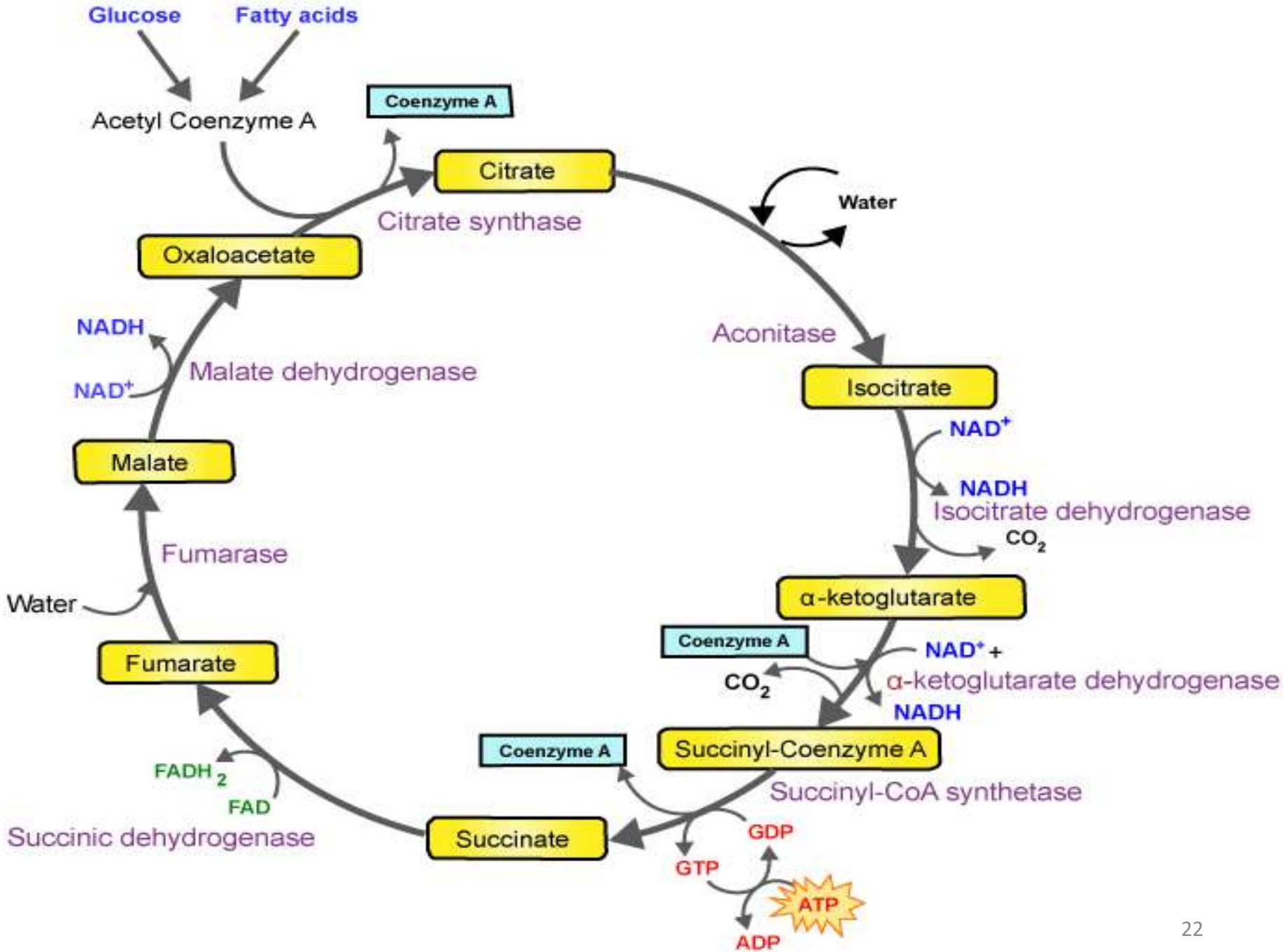
2- Krebs Cycle or Tricarboxylic Acid Cycle (TCA):

The central function is the oxidation of acetyl CoA to CO_2.

- It is the final common pathway for oxidation of fuel molecules.
- Acetyl CoA is derived from the metabolism of fuel molecules as carbohydrates, amino acids, and fatty acids.

- Is an important source of precursors.
- Citric Acid Cycle located in the mitochondrial matrix.

Krebs Cycle


Glucose
Fatty acids
Acetyl Coenzyme A
Coenzyme A
Citrate
Citrate synthase
Water
Oxaloacetate
Aconitase
NADH
NAD+
Malate dehydrogenase
Isocitrate
NAD+
NADH
Isocitrate dehydrogenase
CO2
Malate
Fumarase
α-ketoglutarate
Water
Coenzyme A
NAD+ +
CO2
α-ketoglutarate dehydrogenase
NADH
Fumarate
Coenzyme A
Succinyl-Coenzyme A
FADH2
FAD
Succinyl-CoA synthetase
Succinic dehydrogenase
Succinate
GDP
GTP
ATP
ADP

ATP generated in TCA cycle:

• Isocitric acid to α-ketoglutarate	+1NADH =3ATP
• α –ketoglutarare to succinyl CoA	+1NADH =3ATP
• Succinyl CoA to succinic acid	+1GTP =1ATP
• Succinic acid to fumaric acid	+1FADH2 =2 ATP
• Malic acid to oxaloacetic acid	+1NADH =3ATP
	----------
	total 12 ATP/Pyruvate
	24 ATP/Glucose
• pyruvic acid to acetyl CoA	1NADH = 3ATP/Pyruvate

- Total ATP per glucose (in complete aerobic oxidation) is:

24 (Krebs cycle) + 6ATP (Pyruvate to Acetyl CoA) + 8ATP (Glycolysis)= 38 ATP (in most body tissues)

= 36 ATP (in certain tissues)

Shuttle system in Metabolism

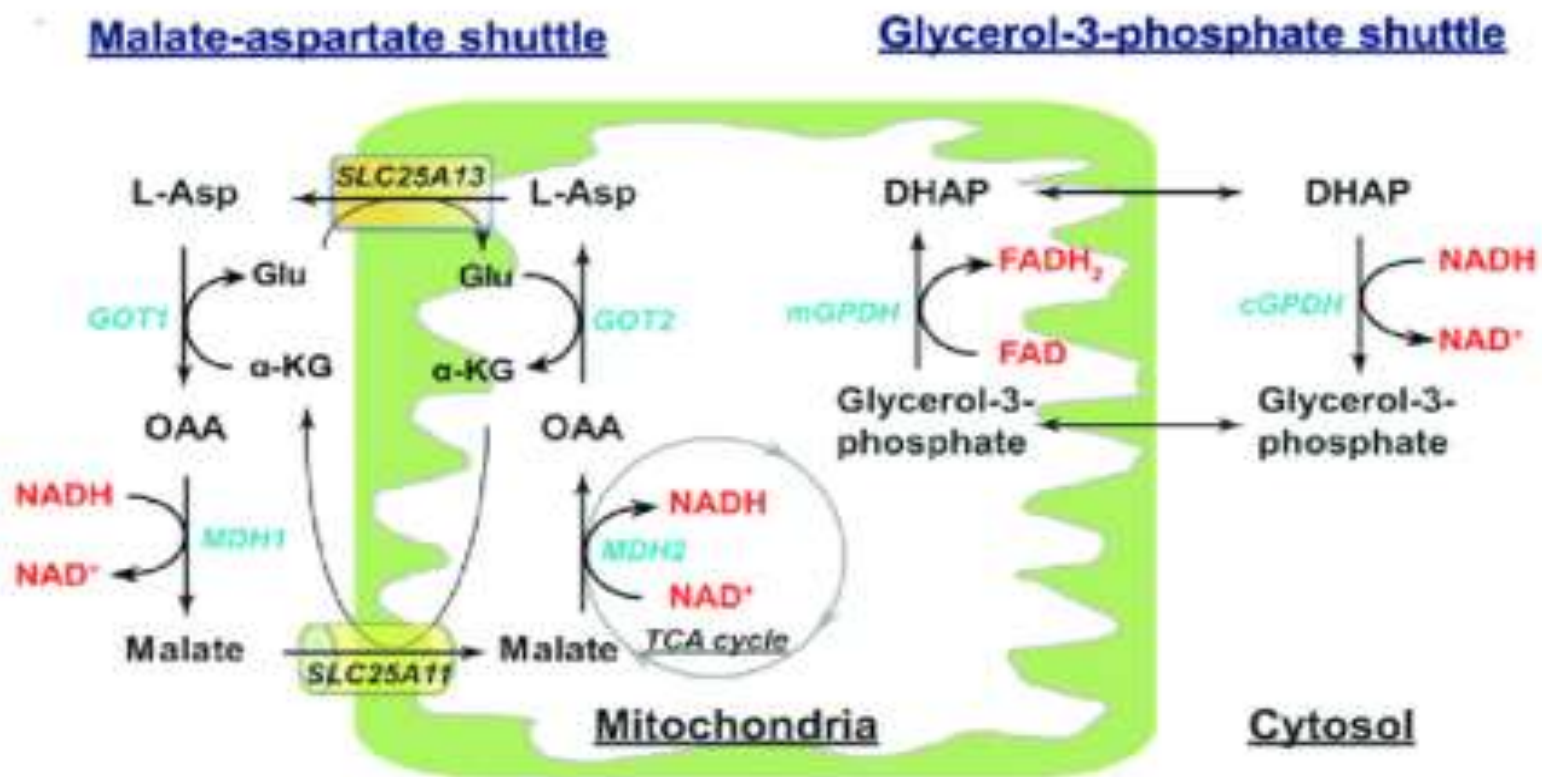
- Shuttle system is a system that helps in the transfer of electrons from NADH to the electron transport chain.

1. **Malate-aspartate Shuttle System.**
2. **Gylcerol 3-Phosphate Shuttle System.**

- This shuttle system is used during glycolysis for the transport of 2NADH (electrons) from cytoplasm to mitochondria matrix for oxidative Phosphorylation.

Shuttle Pathways or Systems

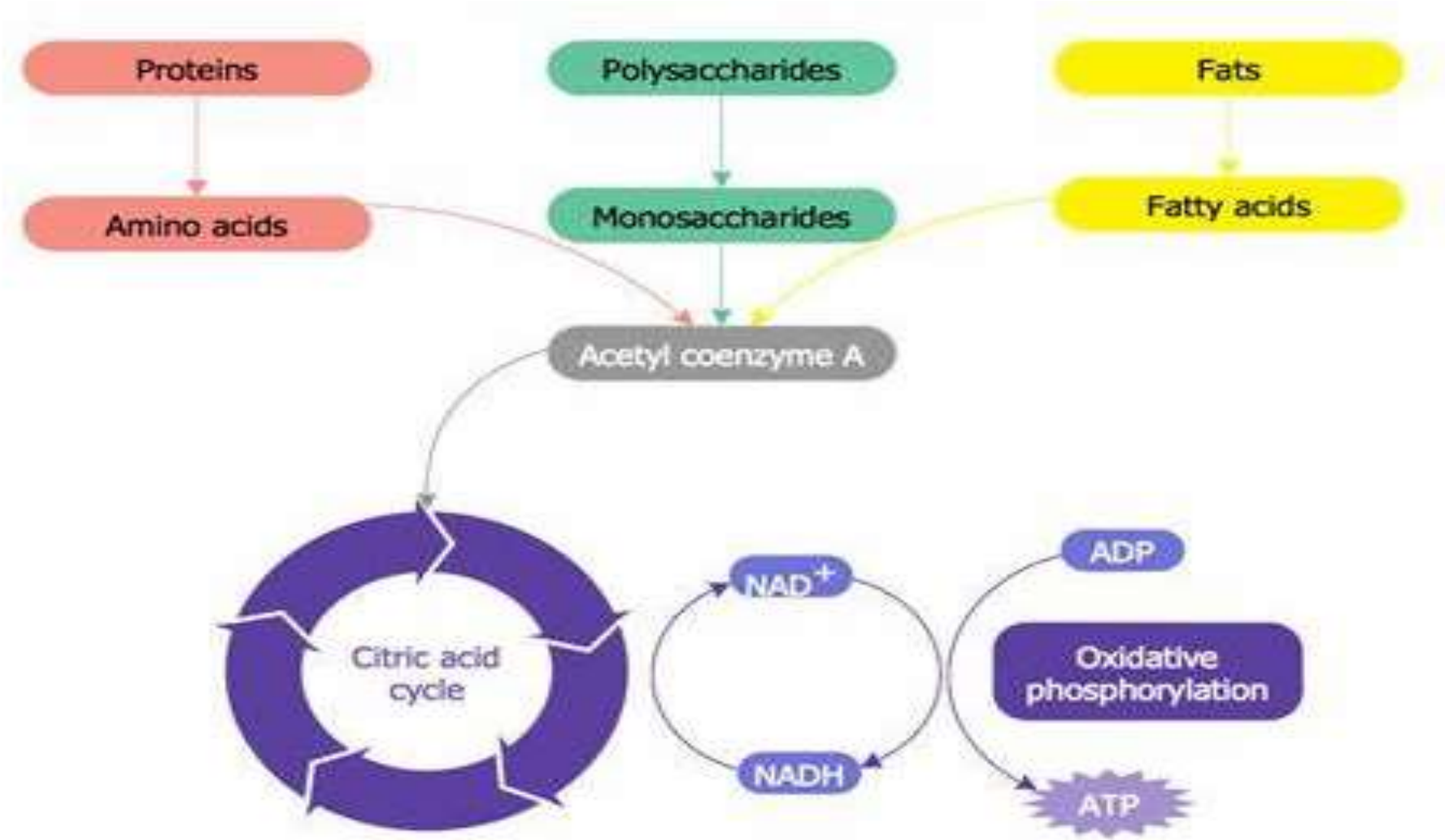
(Transport of reducing equivalents)



- Liver, kidney, and heart utilize malate-aspartate shuttle, and yield 3 ATP per NADH.

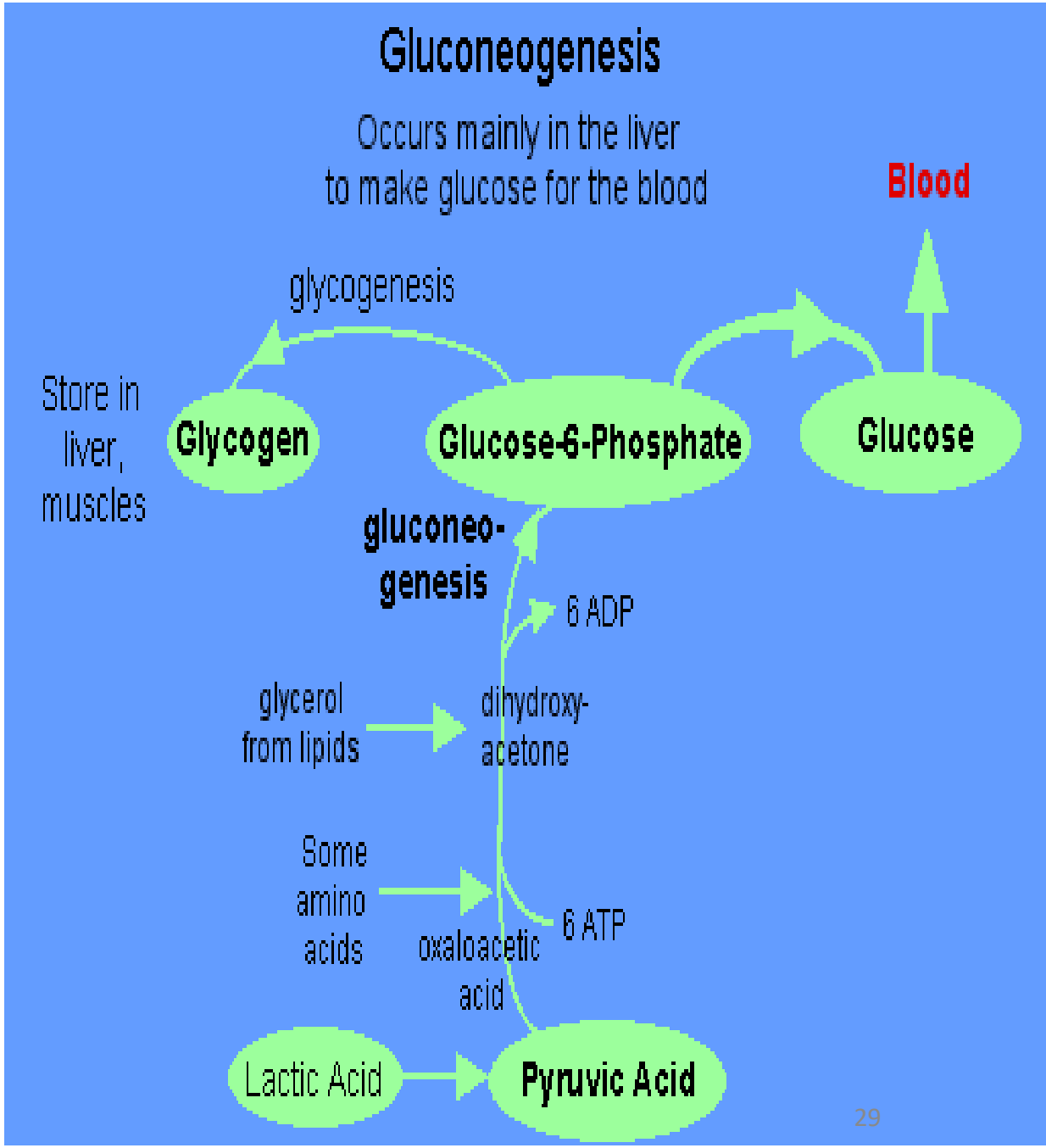
- Skeletal muscle and the brain utilize glycerol-phosphate shuttle and yield 2 ATP from NADH.

Catabolism Schematic



Gluconeogenesis:

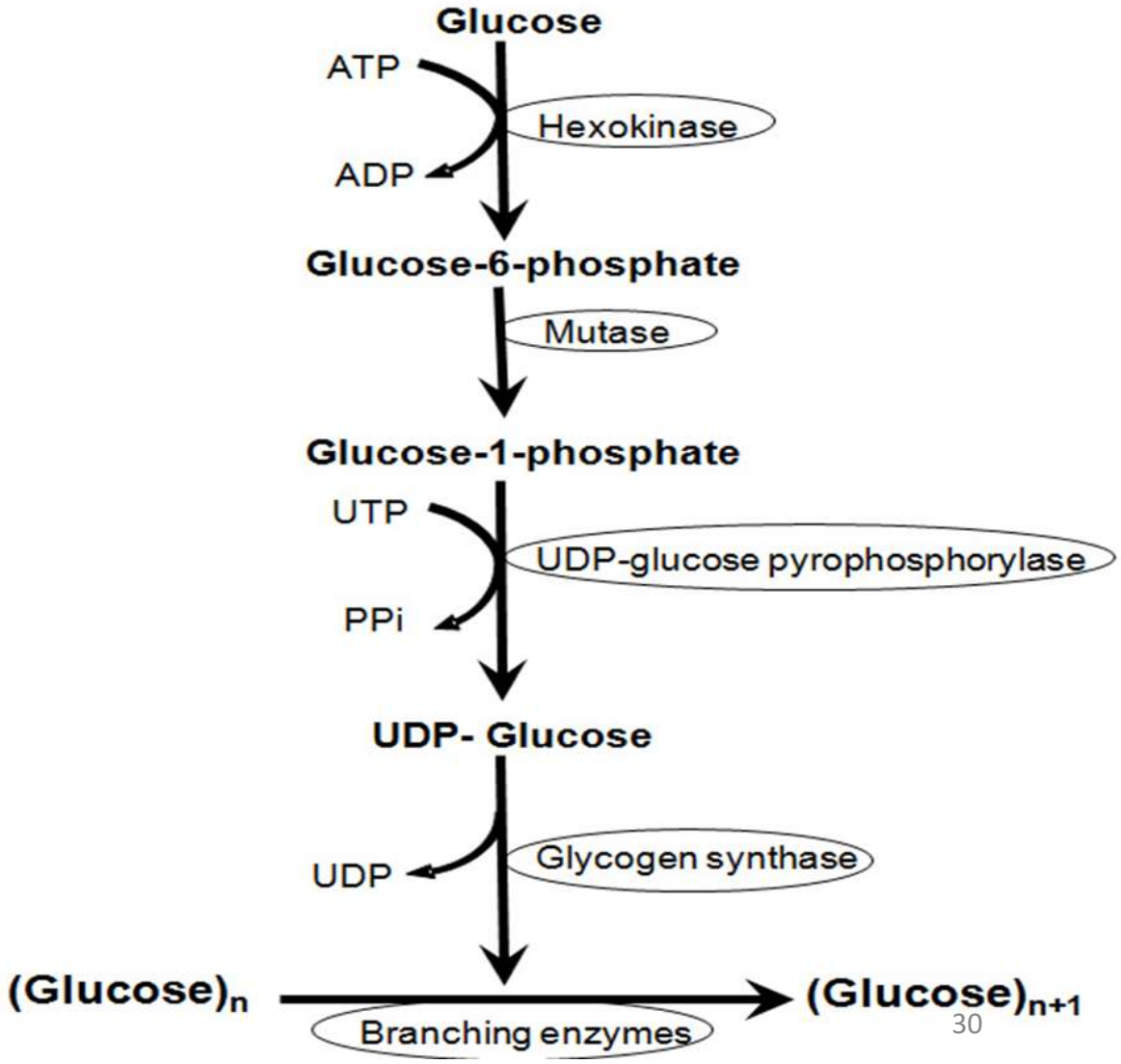
- Is the process of producing glucose from non-carbohydrate sources.
- It mainly occurs in hepatocytes.
- The molecules that provide substrates for gluconeogenesis include proteins, lipids and pyruvate.



Glycogen Metabolism:

1- Glycogenesis (glycogen synthesis):

- Is the formation of glycogen from glucose.
- Glycogen serves as an energy store primarily in muscle and liver.
- It is stored in the form of granules cytoplasm in the cell.



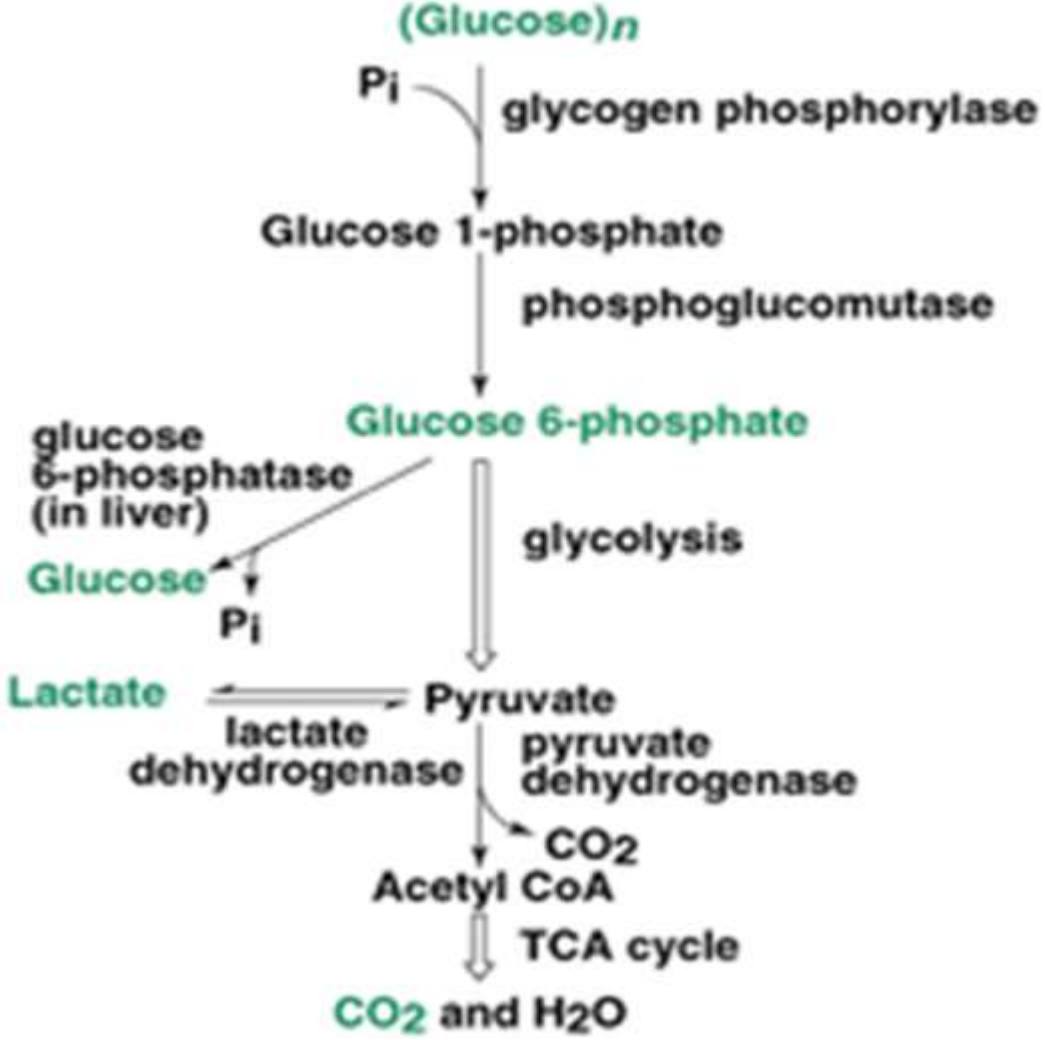
- The concentration of glycogen in muscle is low compared to the levels stored in the liver.

- Glycogen is an energy reserve that can be quickly mobilized to meet a sudden need for glucose.

2- Glycogenolysis (breakdown of glycogen to glucose):

- Take place in the cells of muscle and liver tissues in response to hormonal and neural signals.

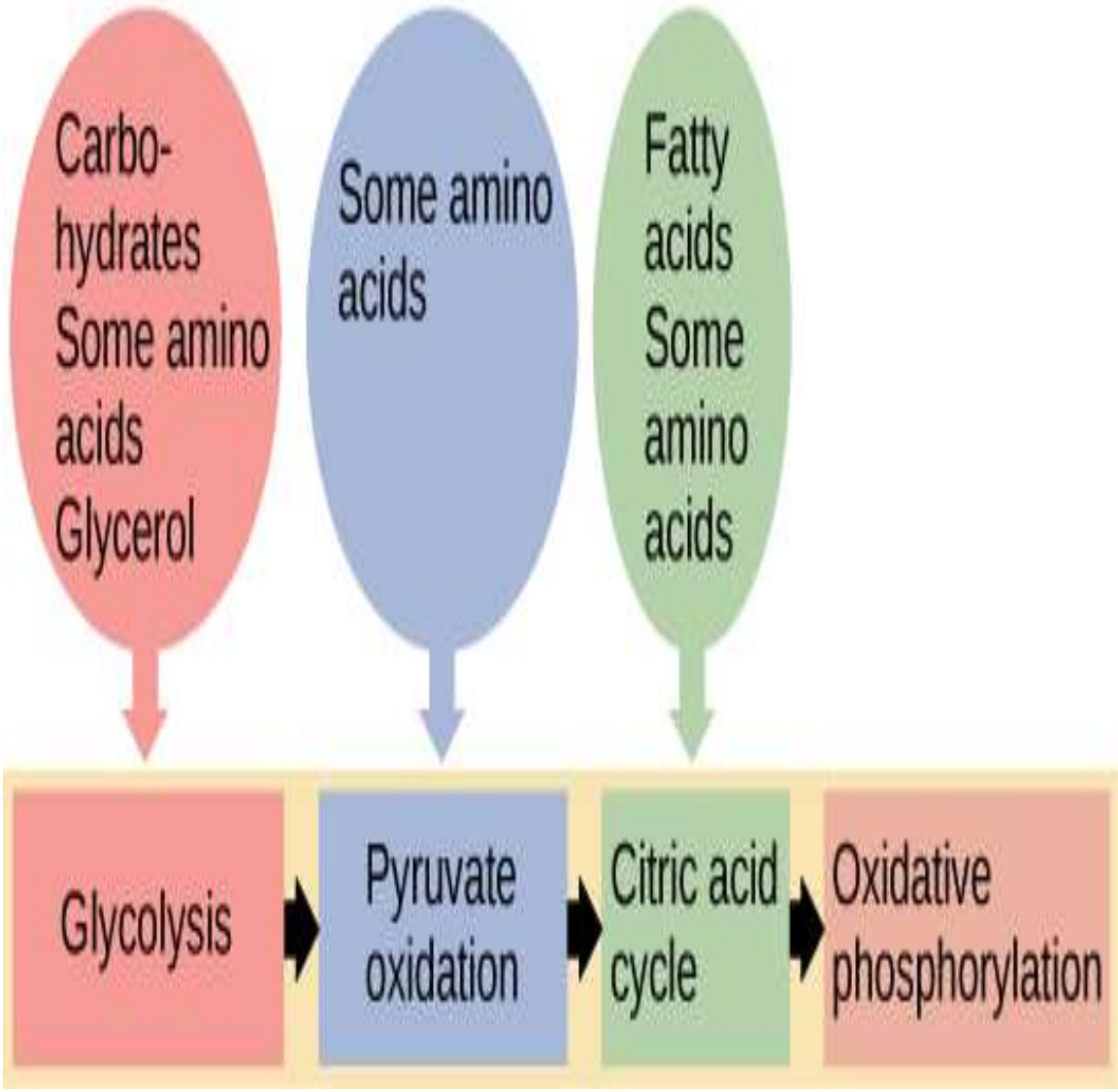
- Glycogenolysis occurs in the cytoplasm and is stimulated by glucagon and adrenaline hormones.



A- Connections of Lipids and Glucose Metabolisms:

- The lipids that are connected to the glucose pathways are cholesterol and triglycerides.

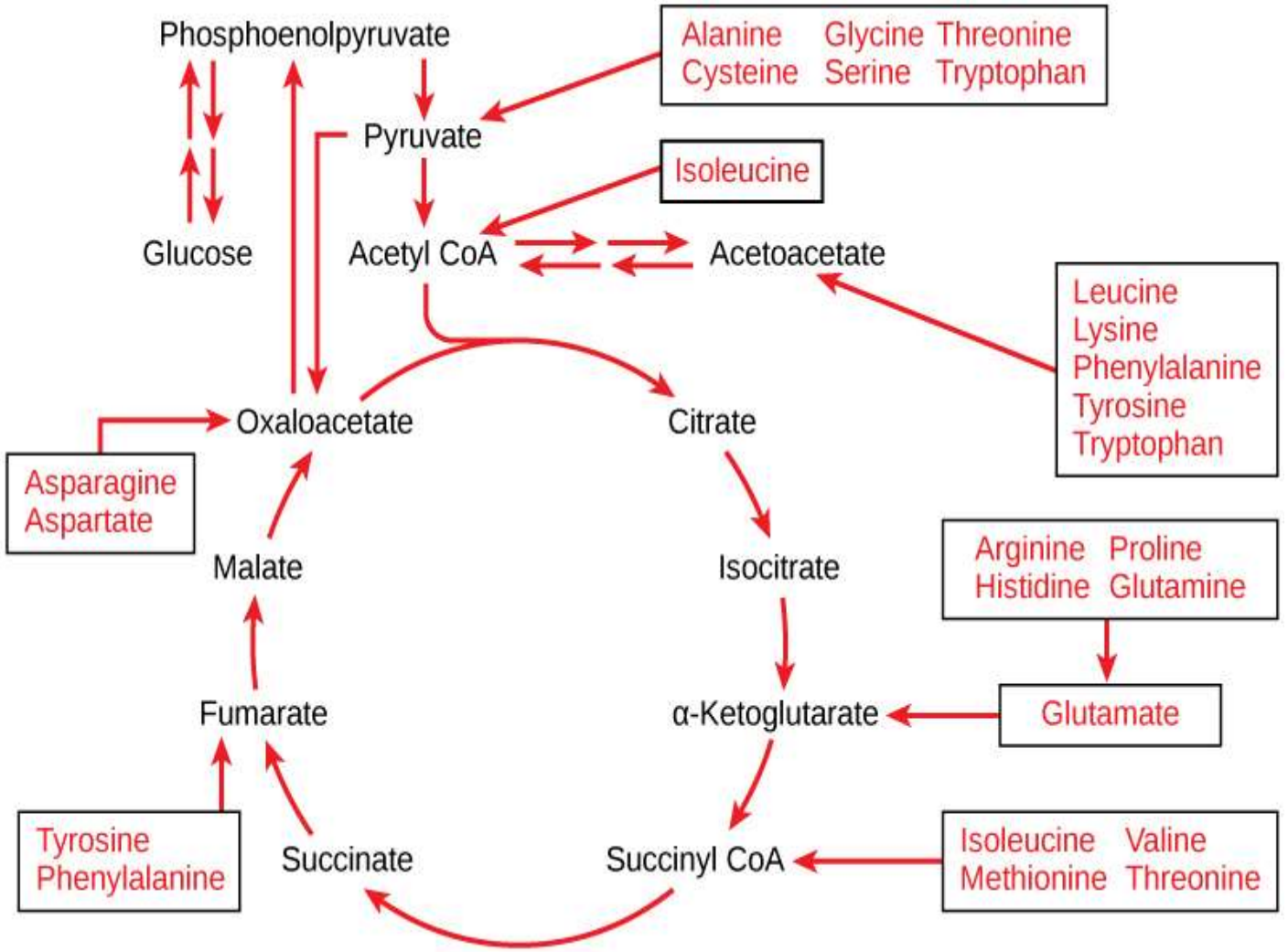
- Glycogen from the liver and muscles, hydrolyzed into glucose-1-p, together with fats and proteins, can feed into the catabolic pathways for carbohydrates.



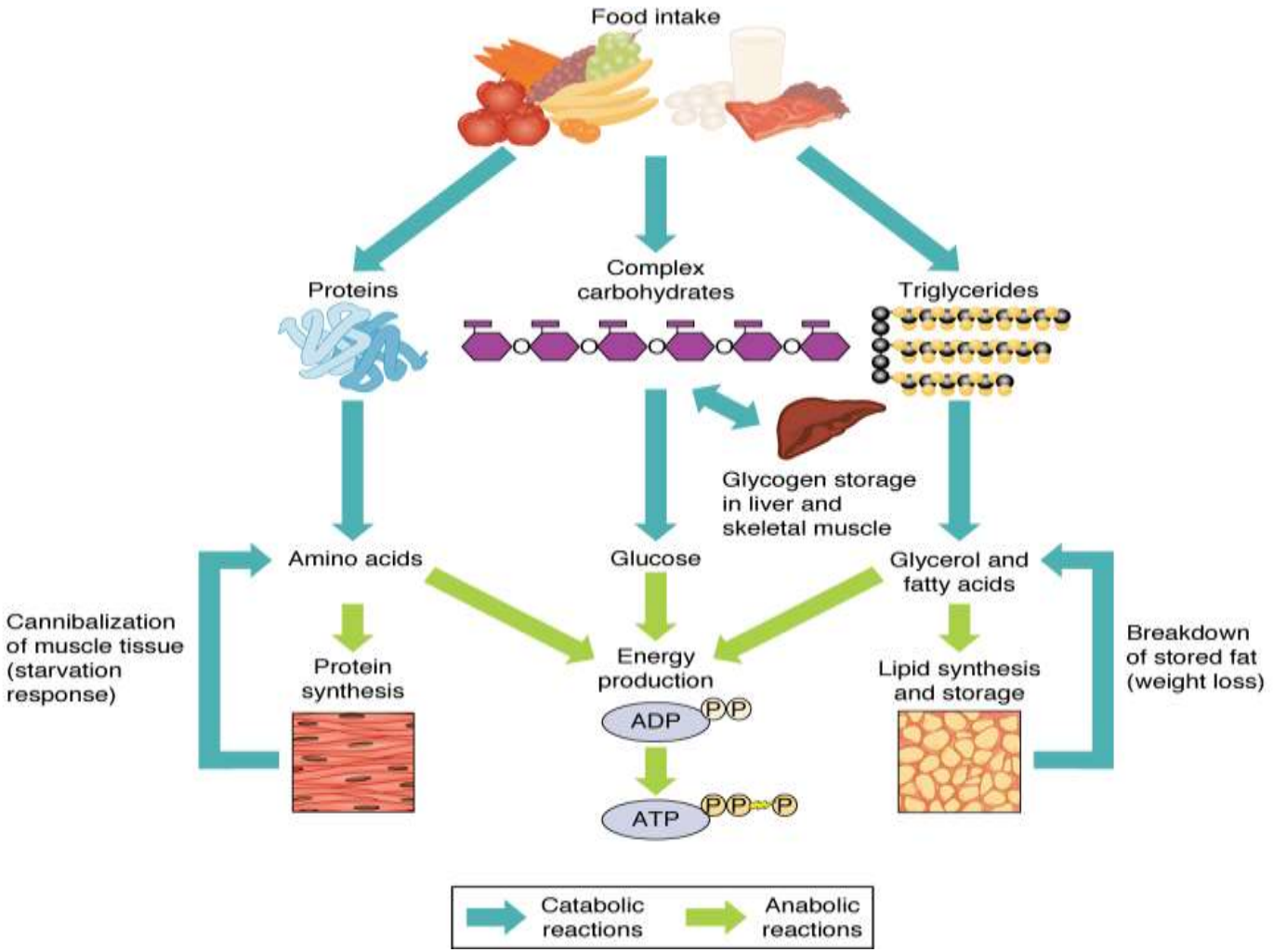
B- Connections of Proteins to Glucose Metabolism:

- Proteins are usually hydrolyzed into a. acids, then a. acids are recycled into the synthesis of new proteins.

- In excess of amino acids, or if the body is in a state of starvation, some amino acids will be shunted into the pathways of glucose catabolism.



Connections of Carbohydrate, Protein, and Lipid Metabolic Pathways



Regulation of blood glucose level

The blood sugar level is maintained by two factors:

a. factors adding glucose to blood (increase blood glucose level).

- from diet (intestinal absorption).
- glycogenolysis (liver).
- gluconeogenesis.
- lipolysis
- conversion of fructose and galactose into glucose.

b. factors remove glucose from blood (decrease blood glucose level).

- glycogen formation in liver and muscle (glycogenesis).
- glycolysis in liver(oxidation of glucose).
- conversion of glucose to fat in adipose tissue(lipogenesis).
- β-oxidation (supply energy).
- synthesis of glycoprotein.
- excretion in urine (diabetes).

Hormones decrease blood glucose level:

- Insulin:

- secrete from β-cells of pancreas.
- causes the liver to convert more glucose into glycogen (glycogenesis).
- about 2/3 of body cells (primarily muscle and fat tissue cells) take up glucose from the blood, thus decreasing blood sugar.

Hormones increase blood glucose level:

1- Glucagon:

- secrete from α-cells of pancreas.
- convert glycogen into glucose (glycogenolysis).

2- Epinephrine:

- also known as adrenalin.
- enhances release of glucose from glycogen (glycogenolysis).
- enhances release of fatty acids from adipose tissue (lipolysis).

3- Cortisol:

- enhances (gluconeogenesis).
- antagonizes Insulin.

4- Thyroxine:

- enhances release of glucose from glycogen.
- enhances absorption of sugars from intestine.

5- ACTH:

- enhances release of fatty acids from adipose tissue (lipolysis).

Blood Glucose Chart			
Mg/DL	Fasting	After Eating	2-3 hours After Eating
Normal	80-100	<180	<130
Impaired Glucose	101-125	190-230	140-160
Diabetic	126+	220-300	200+

Abnormalities in blood glucose level

1-Hyperglycemia:

- is an abnormally high blood glucose level.
- is a hallmark sign of diabetes (both type 1 and type 2 diabetes) and prediabetes.
- Diabetes is the most common cause of hyperglycemia.
- Causes of hyperglycemia: include pancreatitis, Cushing's syndrome, pancreatic cancer, certain medications, …etc.

- Common symptoms of diabetes: polyuria, thirsty, feeling hungry, fatigue, blurry vision, weight loss (type 1), pain or numbness in the hands/feet (type 2), cardiac arrhythmia.
- Treatment: This is done by a combination of proper diet, regular exercise, and insulin or other medication such as metformin.

2- Hypoglycemia:

- abnormally low level of glucose in the blood.
- the brain needs a continuous supply of glucose to function.
- hypoglycemia is not a disease, it is commonly linked with diabetes or caused by other condition.

Hypoglycemia, Cont.

- Causes of hypoglycemia: some liver diseases (hepatitis), Kidney disorders, adrenal and pituitary glands disorders, insulinoma, certain medications, ...etc.
- Common symptoms: include hunger, trembling, heart racing, nausea, and sweating.

Disorders in CHO Metabolism

1- Lactose intolerance:

- is the decreased ability to digest lactose.
- Causes: due to deficiency of lactase enzyme, which may be genetic or acquired.
- Main symptoms: abdominal pain, bloating, diarrhea, gas, and nausea (1/2 -2hrs).

2- Galactosemia:

- it is a hereditary disease.
- Causes: defect in, or absence of, galactose metabolizing enzymes.
- causing the body to be unable to metabolize galactose, so allowing toxic levels of galactose to build up in human body blood, cells, and tissues or urine .

- Symptoms: lethargy, vomiting, diarrhea, failure to thrive, and jaundice.

3-Glycogen storage disease (GSD):

- defects in the processing of glycogen synthesis or breakdown within muscles, and liver.
- causing an accumulation of glycogen in liver or muscle or other tissues.

- Causes: genetic GSD is caused by any inborn error of metabolism (genetic defect of enzymes), or acquired GSD is caused by intoxication.

- There are over 20 types of GSD including the subtypes,
- and are classified based on the deficient enzyme and affected tissues.
- Some examples of these types are:

Types of GSD	Defective enzyme or transporter	Primary tissue involvement
I (von Gierke)	Glucose-6-phosphatase, Glucose-6-phosphatase transporter	Liver
IV (Andersen)	Glycogen branching enzyme	Liver
V (McArdle)	Muscle glycogen phosphorylase	Muscle

References:

1. Rodwell VW, Bender DA, Botham KM, Kennelly PJ, Weil PA. Harper's Illustrated Biochemistry. 30th ed. New York: McGraw-Hill Education; 2015. Chapter 17, Glycolysis & the Oxidation of Pyruvate, p. 168-175.
2. Ferguson AM. Carbohydrate disorders. In: Garg U, Smith LD, eds. Biomarkers in Inborn Errors of Metabolism: Clinical Aspects and Laboratory Determination. Elsevier; 2017: 155-166.
3. Gümüş E, Özen H. Glycogen storage diseases: An update. World J Gastroenterol. 2023 Jul 7;29(25):3932-3963. doi: 10.3748/wjg.v29.i25.3932.
4. Murray RK, Bender DA, Botham KM, et al. Harper's Illustrated Biochemistry. 31st ed. New York: McGraw-Hill Education; 2018.


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